⚛️ Honors Physics 11 — Complete Review Sheet

All key concepts, formulas, and definitions for the 1st & 2nd Trimester exams. Know your units, show your work, and use the reference table.

📝 1st Tri: Chapters 1–7 | 📝 2nd Tri: Units 6–10 | 📐 SI Units throughout | 🖨️ Print-friendly

⚡ Quick Reference — 1st Trimester Key Formulas

AVG VELOCITY $v = \Delta d / \Delta t$	ACCELERATION $a = \Delta v / \Delta t$
KINEMATIC #1 $v_f = v_i + at$	KINEMATIC #2 $d = v_i \cdot t + \frac{1}{2}at^2$
KINEMATIC #3 $v_f^2 = v_i^2 + 2ad$	NEWTON'S 2ND $F_{net} = ma$
FRICTION $f = \mu N$	WORK $W = Fd \cdot \cos\theta$
KINETIC ENERGY $KE = \frac{1}{2}mv^2$	GRAV. PE $PE_g = mgh$
SPRING PE $PE_s = \frac{1}{2}kx^2$	POWER $P = W/t = Fv$
MOMENTUM $p = mv$	IMPULSE $J = F\Delta t = \Delta p$

CH 1

Introduction to Physics & Math Review

~2-3 multiple choice questions

KEY CONCEPTS

- Scientific notation
- Significant figures
- SI units (m, kg, s, N, J, W)
- Unit conversions
- Graphing: slope, y-intercept
- Direct vs inverse relationships
- Proportional reasoning
- Solving equations algebraically

📐 **Graphing tip:** The slope of a graph = the variable on the y-axis divided by the variable on the x-axis. A straight line means a direct (linear) relationship; a curve through origin often means direct square.

CH 2

Describing Motion

~8 multiple choice questions

DEFINITIONS

Scalar
Magnitude only (e.g. speed, distance, time)

Vector
Magnitude + direction (e.g. velocity, displacement)

Distance
Total path length traveled — scalar (m)

Displacement (Δd)
Straight-line change in position — vector (m). $\Delta d = d_f - d_i$

Speed
Distance / time — scalar (m/s)

Velocity
Displacement / time — vector (m/s)

Average vs Instantaneous
Average = over whole trip; Instantaneous = at one moment (slope of tangent)

FORMULAS

AVERAGE VELOCITY

$$v_{avg} = \Delta d / \Delta t$$

$= (d_f - d_i) / (t_f - t_i)$

AVERAGE SPEED

$$\text{speed} = \text{total distance} / \text{total time}$$

GRAPH INTERPRETATIONS

d-t graph slope

= velocity (steep = fast, flat = stopped)

v-t graph slope

= acceleration

v-t graph area

= displacement

🔑 **Vector addition:** Draw displacement vectors tip-to-tail. The resultant goes from start to final tip. For perpendicular vectors: $|R| = \sqrt{(A^2 + B^2)}$, angle = $\arctan(B/A)$.

CH 3

Acceleration

~8 multiple choice + 2 calculation questions

KINEMATIC EQUATIONS (USE WHEN ACCELERATION IS CONSTANT)

Equation	Formula	Missing Variable
Eq. 1	$v_f = v_i + at$	d
Eq. 2	$d = v_i \cdot t + \frac{1}{2}at^2$	v_f
Eq. 3	$v_f^2 = v_i^2 + 2ad$	t
Eq. 4	$d = [(v_i + v_f)/2] \cdot t$	a

FREE FALL

FREE FALL ACCELERATION

$g = 9.8\ m/s^2 \downarrow$

Downward = negative in most sign conventions

ALL OBJECTS FALL EQUALLY

mass doesn't matter (ignore air resistance)

VERTICAL LAUNCH (THROWN STRAIGHT UP)

AT MAX HEIGHT (APEX)

$v_y = 0\ m/s$

Momentarily stopped going up, acceleration still 9.8 $m/s^2 \downarrow$

TIME TO MAX HEIGHT

$t_{up} = v_i / g$

MAX HEIGHT

$h_{max} = v_i^2 / (2g)$

HANG TIME (SYMMETRY)

$t_{hang} = 2 \times t_{up}$

Up time = down time; symmetrical

⚡ **Key graph fact:** Slope of a v-t graph = acceleration. A straight line on a v-t graph means constant acceleration. Area under v-t graph = displacement.

CH 4

Vectors & Projectile Motion

~8 multiple choice + 2 calculation questions

VECTOR COMPONENTS

X-COMPONENT

$A_x = A \cdot \cos\theta$

Y-COMPONENT

$A_y = A \cdot \sin\theta$

MAGNITUDE FROM COMPONENTS

$|A| = \sqrt{A_x^2 + A_y^2}$

DIRECTION ANGLE

$\theta = \arctan(A_y / A_x)$

PROJECTILE MOTION RULES

Horizontal (x)

Constant velocity: $a_x = 0$

$x = v_x \cdot t$, where $v_x = v_i \cdot \cos\theta$

Vertical (y)

Constant acceleration: $a_y = -g$

$v_{iy} = v_i \cdot \sin\theta$

Independence

Horizontal and vertical motions are completely independent — solve separately!

PROJECTILE KEY POINTS

AT MAX HEIGHT (APEX)

$v_y = 0$

v_x remains constant throughout flight

HANG TIME

use vertical: $2 \times (v_{iy} / g)$

RANGE

LAUNCHED HORIZONTALLY

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x = v_x · t_hang
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v_iy = 0; only v_x initial
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🎯 **Projectile problem strategy:** (1) Break initial velocity into v_x and v_y components. (2) Use vertical kinematics to find time. (3) Use horizontal kinematics to find range or position. Time is the link between x and y!

CH 5

Forces

~10 multiple choice + 2 calculation questions

NEWTON'S THREE LAWS

1st Law — Inertia

An object stays at rest or in constant motion unless a NET force acts on it. If $F_{net} = 0 \rightarrow a = 0$.

2nd Law — F = ma

Net force = mass × acceleration. Direction of acceleration = direction of net force. Units: N = $kg \cdot m/s^2$

3rd Law — Action-Reaction

For every action there is an equal and opposite reaction. Forces act on DIFFERENT objects.

KEY FORCES & FORMULAS

NET FORCE

$$F_{net} = ma$$

[N = $kg \cdot m/s^2$]

WEIGHT

$$F_W = mg$$

g = 9.8 m/s^2; weight acts downward

KINETIC FRICTION

$$f_k = \mu_k \cdot N$$

Opposes motion; μ_k always < μ_s

STATIC FRICTION (MAX)

$$f_s \leq \mu_s \cdot N$$

Before object moves; can be less than max

INCLINES & SPECIAL CASES

INCLINE: PARALLEL FORCE

$$F_{\parallel} = mg \cdot \sin\theta$$

Acts down the slope

INCLINE: PERP. FORCE

$F_{\perp} = mg \cdot \cos\theta = N$

Normal force on frictionless incline

ELEVATOR: GOING UP/ACCEL

$F_N = m(g + a)$

Feel heavier (apparent weight > real)

ELEVATOR: GOING DOWN/ACCEL

$F_N = m(g - a)$

Feel lighter; $F_N = 0$ in free fall (weightless)

📊 **Free Body Diagrams:** Always draw ALL forces on the object — Weight (down), Normal (⊥ surface), Friction (opposing motion), Applied forces, Tension. F_{net} = sum of ALL forces.

🚀 **Terminal velocity:** When drag force = gravity ($F_{drag} = F_W$), acceleration = 0 and speed is constant. Air resistance increases with speed until equilibrium.

CH 6

Work, Energy & Power

~10 multiple choice + 2 calculation questions

FORMULAS

WORK

$$W = F \cdot d \cdot \cos\theta$$

θ = angle between force & displacement. Units: Joules (J = N·m). Zero work if F ⊥ d (θ = 90°)

KINETIC ENERGY

$$KE = \frac{1}{2}mv^2$$

Energy of motion [J]. Doubles when v doubles? → KE × 4!

GRAVITATIONAL PE

$$PE_g = mgh$$

h = height above reference [J]

SPRING PE (ELASTIC)

$$PE_s = \frac{1}{2}kx^2$$

k = spring constant (N/m), x = compression/stretch

WORK-ENERGY THEOREM

$$W_{net} = \Delta KE = KE_f - KE_i$$

Net work done = change in KE

POWER

$$P = W / t = F \cdot v$$

Units: Watts (W = J/s)

CONSERVATION OF ENERGY

NO FRICTION (IDEAL)

$$KE_i + PE_i = KE_f + PE_f$$

Total mechanical energy stays constant

WITH FRICTION

$$KE_i + PE_i = KE_f + PE_f + Q$$

Q = heat/internal energy lost to friction

MECHANICAL ENERGY

$$ME = KE + PE$$

FORMS OF ENERGY

- Kinetic (KE)
- Gravitational PE
- Elastic PE (springs)
- Thermal/Internal
- Chemical
- Nuclear
- Electrical

💡 **Energy conservation strategy:** Set PE = 0 at the lowest point. At the top: KE = 0 (if starting from rest), PE = mgh. At the bottom: PE = 0, all energy is KE. Use KE_top + PE_top = KE_bottom + PE_bottom.

CH 7

Momentum & Impulse

~8 multiple choice + 2 calculation questions

FORMULAS

MOMENTUM

$$p = mv$$

Vector! Units: kg·m/s

IMPULSE

$$J = F \cdot \Delta t = \Delta p = m\Delta v$$

Units: N·s = kg·m/s

CONSERVATION OF MOMENTUM

$$p_{before} = p_{after}$$

$m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$

PERFECTLY INELASTIC

$$(m_1+m_2)v_f = m_1v_1 + m_2v_2$$

Objects stick together after collision

COLLISION TYPES

Elastic Collision

Momentum AND kinetic energy both conserved. Objects bounce. (e.g., billiard balls)

Inelastic Collision

Momentum conserved; KE NOT fully conserved — some lost as heat/sound/deformation.

Perfectly Inelastic

Objects stick together. Maximum KE loss. Momentum still conserved.

Explosion

Objects start together, fly apart. Total momentum stays = 0 if initially at rest.

🚗 **Airbags & crumple zones:** Increase the time (Δt) of impact → same impulse (Δp) over longer time → smaller force (F = Δp/Δt). This is why longer stop times = smaller forces =

> safer crashes.

⚡ Quick Reference — 2nd Trimester Key Formulas

ANGULAR VELOCITY
$\omega = \Delta\theta / \Delta t$

TORQUE
$\tau = r \cdot F \cdot \sin\theta$

ROTATIONAL KE
$KE_{rot} = \frac{1}{2} I\omega^2$

ANGULAR MOMENTUM
$L = I\omega$

UNIVERSAL GRAVITY
$F_g = Gm_1m_2/r^2$

CENTRIPETAL ACCEL
$a_c = v^2/r$

CENTRIPETAL FORCE
$F_c = mv^2/r$

DENSITY
$\rho = m/V$

PRESSURE
$P = F/A$

BUOYANCY
$F_b = \rho_f \cdot V_{sub} \cdot g$

CONTINUITY
$A_1v_1 = A_2v_2$

BERNOULLI
$P + \frac{1}{2}\rho v^2 + \rho gh = const$

U6 ★ Energy, Work & Power — Cumulative Review

Reviewed from 1st Tri — 6–10 multiple choice questions

★ **This topic carries over from 1st Tri.** Review Ch 6 on the 1st Trimester tab for the full formula set. Key additions for 2nd Tri: internal energy (heat Q), understanding energy transformations in rotational systems.

CONSERVATION (WITH HEAT)

$KE_i + PE_i = KE_f + PE_f + Q$

TOTAL MECHANICAL ENERGY

$ME = KE + PE_g + PE_s$

POWER

$P = W/t = F \cdot v$ [Watts]

WORK-ENERGY THEOREM

$W_{net} = \Delta KE$

U7 Linear Momentum

6–10 multiple choice + 2 calculation questions

★ **Deeper than 1st Tri Ch 7.** Be especially clear on the Impulse-Momentum Theorem and applying conservation to explosions.

MOMENTUM

$p = mv$ [kg·m/s]

IMPULSE-MOMENTUM THEOREM

$J = F\Delta t = \Delta p = m(v_f - v_i)$

CONSERVATION OF MOMENTUM

$\Sigma p_{before} = \Sigma p_{after}$

PERFECTLY INELASTIC

$(m_1+m_2)v_f = m_1v_1 + m_2v_2$

EXPLOSION (FROM REST)

$m_1v_1' + m_2v_2' = 0$

Vectors! Opposite directions

🚗 **Airbags:** Increase Δt → same impulse but smaller force. F = Δp/Δt — longer time = safer impact.

U8 Rotational Motion & Equilibrium

6–10 multiple choice + 2 calculation questions

ANGULAR QUANTITIES

ANGULAR POSITION

θ [radians]

1 rev = 2π rad = 360°

ARC LENGTH

$s = r \cdot \theta$

θ must be in radians

ANGULAR VELOCITY

$\omega = \Delta\theta / \Delta t$ [rad/s]

$\omega = 2\pi f = 2\pi/T$

RPM → RAD/S

$\omega = RPM \times (2\pi/60)$

ANGULAR ACCELERATION

$\alpha = \Delta\omega / \Delta t$ [rad/s^2]

LINEAR ↔ ANGULAR

$v = r\omega$; $a_t = r\alpha$

Linear tangential speed/accel at radius r

TORQUE, INERTIA & ANGULAR MOMENTUM

TORQUE

$\tau = r \cdot F \cdot \sin\theta$ [N·m]

r = distance from pivot; θ = angle between r and F

ROTATIONAL NEWTON'S 2ND

$\tau_{net} = I \cdot \alpha$

MOMENT OF INERTIA — HOOP

$I = mr^2$

All mass at edge (ring, thin-walled cylinder)

MOMENT OF INERTIA — DISK

$I = \frac{1}{2}mr^2$

Solid disk/cylinder — disk is easier to spin!

ROTATIONAL KE

$KE_{rot} = \frac{1}{2}I\omega^2$

ANGULAR MOMENTUM

$L = I\omega$ [kg·m²/s]

CONSERVATION OF L

$I_1\omega_1 = I_2\omega_2$

Spin faster = smaller I (tuck in arms)

🌀 **Hoop vs Disk:** A hoop has MORE inertia ($I = mr^2$) than a disk ($I = \frac{1}{2}mr^2$) of the same mass and radius. The disk wins a race rolling downhill — it's easier to get spinning. Also: rolling objects have BOTH translational and rotational KE.

U9A

Gravity

6–10 multiple choice + 2 calculation questions (shared with UCM)

UNIVERSAL GRAVITATION

LAW OF UNIVERSAL GRAVITATION

$$F_g = G \cdot m_1 \cdot m_2 / r^2$$

$G = 6.67 \times 10^{-11}$ N·m²/kg² (Universal Gravitational Constant — Big G)

INVERSE SQUARE LAW

$$\text{If } r \times 2 \rightarrow F_g \div 4$$

Double distance = force becomes ¼. Triple → 1/9

GRAVITATIONAL FIELD STRENGTH

$$g = F_g / m = GM / r^2$$

Units: m/s² = N/kg; varies by planet/altitude

WEIGHT FROM GRAVITY

$$F_W = mg$$

$g \approx 9.8$ m/s² at Earth's surface

KEY PEOPLE & CONCEPTS

Newton

Formulated universal gravitation law. Gravity is a force between any two masses.

Cavendish

First measured Big G experimentally using a torsion balance (1798).

Gravitational field lines

Point toward the mass; stronger (closer together) near the mass surface.

Difference: g vs G

Little g = local acceleration (9.8 m/s² on Earth). Big G = universal constant (6.67×10^{-11}).

U9B

Uniform Circular Motion (UCM)

6–10 multiple choice + 2 calculation questions (shared with Gravity)

FORMULAS

UCM SPEED

$$v = 2\pi r / T$$

T = period (time for 1 revolution)

CENTRIPETAL ACCELERATION

$$a_c = v^2 / r$$

Always directed toward center (centripetal = "center-seeking")

CENTRIPETAL FORCE

$$F_c = ma_c = mv^2 / r$$

NOT a new force — it's the NET inward force

VECTOR DIRECTIONS & WHAT PROVIDES F_C

Velocity

Tangent to the circle (always perpendicular to radius). Changes direction constantly.

Acceleration & Force

Always point toward the center. If the inward force stops → object moves in a straight line (tangent).

Gravity provides F_c

Planets orbiting the sun, satellites orbiting Earth — gravity acts as the centripetal force.

Tension provides F_c

Ball on a string swung in a horizontal circle.

Normal force provides F_c

Car going over a hill or through a loop-the-loop (inside top).

Friction provides F_c

Car turning on a flat road.

⭕ **Key insight:** Centripetal force is NOT a separate force — it's the name we give to whatever net inward force keeps the object in a circle. Identify which existing force (tension, gravity, normal, friction) is acting inward, set it equal to mv^2/r, and solve.

U10 Fluids

6–10 multiple choice + 2 calculation questions

HYDROSTATICS (FLUIDS AT REST)

DENSITY

$\rho = m / V$ [kg/m³]

Water: $\rho = 1000$ kg/m³; Air: $\rho \approx 1.2$ kg/m³

PRESSURE

$P = F / A$ [Pa = N/m²]

GAUGE PRESSURE

$P_{gauge} = \rho g h$

Pressure due to fluid depth only

ABSOLUTE PRESSURE

$P = P_{atm} + \rho g h$

$P_{atm} \approx 101{,}325$ Pa ≈ 101.3 kPa

BUOYANT FORCE (ARCHIMEDES)

$F_b = \rho_{fluid} \cdot V_{sub} \cdot g$

= weight of fluid displaced

FLOATING CONDITION

$F_b = F_{weight} \rightarrow \rho_{obj} < \rho_{fluid}$

PASCAL'S PRINCIPLE & HYDRAULICS

PASCAL'S PRINCIPLE

Pressure transmitted equally throughout closed fluid

HYDRAULICS

$F_1/A_1 = F_2/A_2$

Small force on small area → large force on large area

HYDRODYNAMICS (FLUIDS IN MOTION)

VOLUME FLOW RATE

$Q = A \cdot v$ [m^3/s]

MASS FLOW RATE

$\dot{m} = \rho \cdot A \cdot v$ [kg/s]

CONTINUITY EQUATION

$A_1 v_1 = A_2 v_2$

Narrow pipe → faster fluid (same flow rate)

BERNOULLI'S EQUATION

$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$

Faster flow = lower pressure (Bernoulli effect)

TORRICELLI'S THEOREM

$v = \sqrt{2gh}$

Speed of fluid exiting a hole at depth h

🌊 **Bernoulli intuition:** Faster-moving fluid has LOWER pressure. This explains airplane lift (faster air over curved top → lower pressure above wing), carburetors, and the curve of a spinning baseball. Continuity: when a pipe narrows, the fluid MUST speed up to maintain the same flow rate.

🪣 **Archimedes' key idea:** Whether an object floats or sinks depends entirely on whether ρ_object > or < ρ_fluid. The buoyant force = weight of fluid displaced (not weight of object). Even a heavy steel ship floats because its average density (hull + air inside) < water.

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